

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

Duration :60 Minutes

Total Marks: 40

Code of Conduct

I Atmakuri Sai Mrudula/192372087 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Mrudula

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- | | | | |
|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |
- | | | | |
|--|--------------|----------------|----------------|
| Agriculture | Smart | Farming | Problem |
| A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated | | | |

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

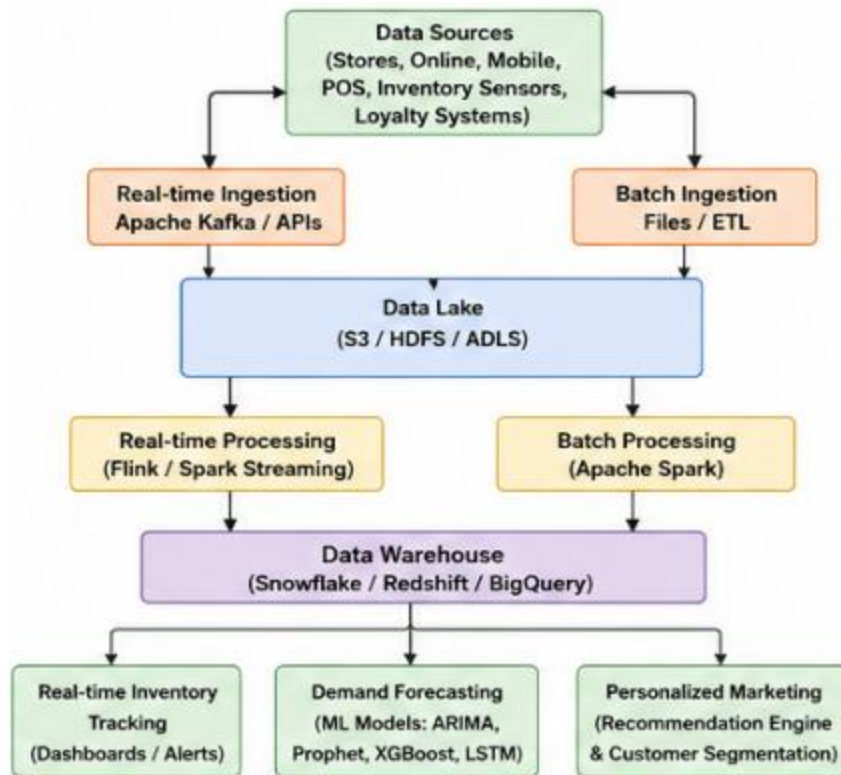
Total Marks Awarded:

- 1. Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

The retail chain generates data from physical stores, online shopping, inventory systems, customer transactions, mobile applications, and customer behavior. The main requirements are:

- Real-time inventory tracking
- Demand forecasting
- Personalized marketing
- Integration of online and offline data
- Scalable processing of large datasets

Proposed Architecture: -



Tools and Technologies: -

1. Data Sources

- Point-of-Sale systems
- E-commerce website
- Mobile applications
- RFID and inventory sensors
- Customer loyalty systems

2. Data Ingestion

Apache Kafka can collect continuous sales and inventory events.

For example:

Product sold → POS → Kafka → Processing → Inventory updated

3. Data Storage

A cloud data lake such as Amazon S3, Azure Data Lake, or Google Cloud Storage can store large volumes of structured and unstructured data.

4. Data Processing

Apache Spark can process historical data using batch processing. Apache Flink or Spark Structured Streaming can process real-time inventory events.

Analytics Models: -

Demand Forecasting

Time-series and machine learning models such as:

- ARIMA
- Prophet
- Random Forest
- XGBoost
- LSTM

can predict future product demand.

Personalized Recommendations

Collaborative filtering and content-based recommendation can analyze:

- Purchase history
- Browsing history
- Product preferences
- Similar customer behavior

Benefits

- Real-time inventory visibility
- Reduced stockouts
- Reduced overstock
- Better demand forecasting
- Personalized marketing
- Improved customer satisfaction

A combination of Kafka, cloud data lakes, Spark/Flink, machine learning, and recommendation algorithms provides a scalable architecture for real-time inventory management and personalized retail analytics.

2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

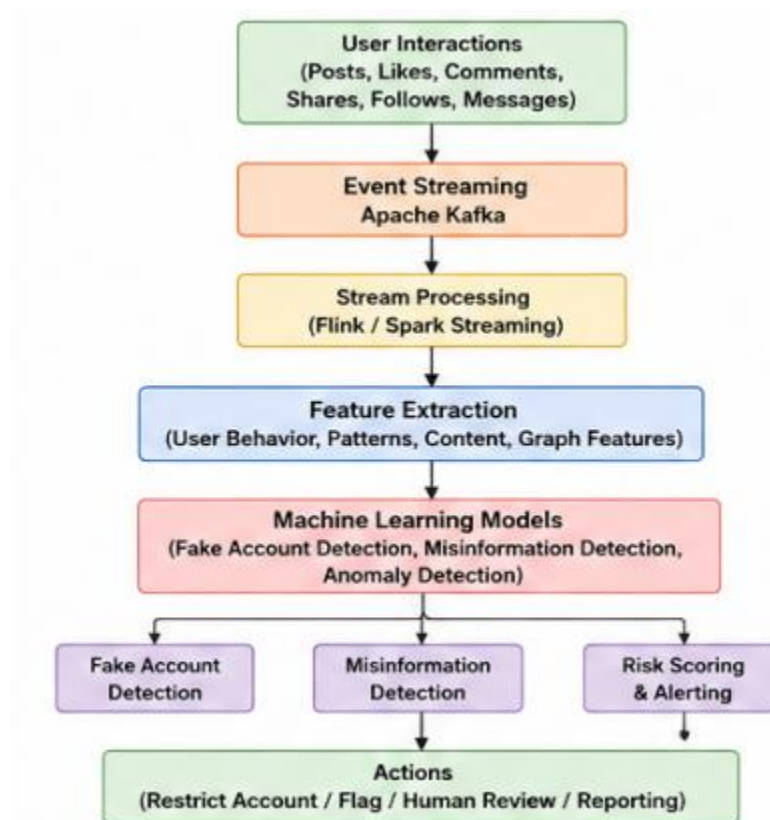
A social media platform generates billions of:

- Posts
- Comments
- Likes
- Shares
- Follows
- Messages

The platform must quickly detect:

- Fake accounts
- Spam
- Coordinated activity
- Misinformation
- Abnormal user behavior

Because the data arrives continuously, a streaming architecture is required.



Machine Learning for Fake Account Detection

Features can include:

- Account age
- Number of followers
- Following/follower ratio
- Posting frequency
- Login locations
- IP/device patterns
- Similarity with other accounts
- Repeated content

Algorithms:

- Random Forest
- XGBoost
- Neural Networks
- Isolation Forest for anomaly detection

Misinformation Detection: -

Natural Language Processing can analyze:

- Text
- Keywords
- Sentiment
- Content similarity
- Source reliability
- Propagation patterns

Graph analytics can identify coordinated groups of accounts sharing the same content.

Batch Processing

Historical data can be processed using Spark to identify long-term patterns and retrain ML models.

Benefits: -

- Near real-time detection
- Handles billions of events
- Scalable processing
- Continuous model improvement
- Detection of coordinated fraudulent behavior
- Reduced manual monitoring

A combination of Kafka, Flink/Spark Streaming, machine learning, NLP, and graph analytics provides a scalable solution for detecting fake accounts and misinformation with low latency.

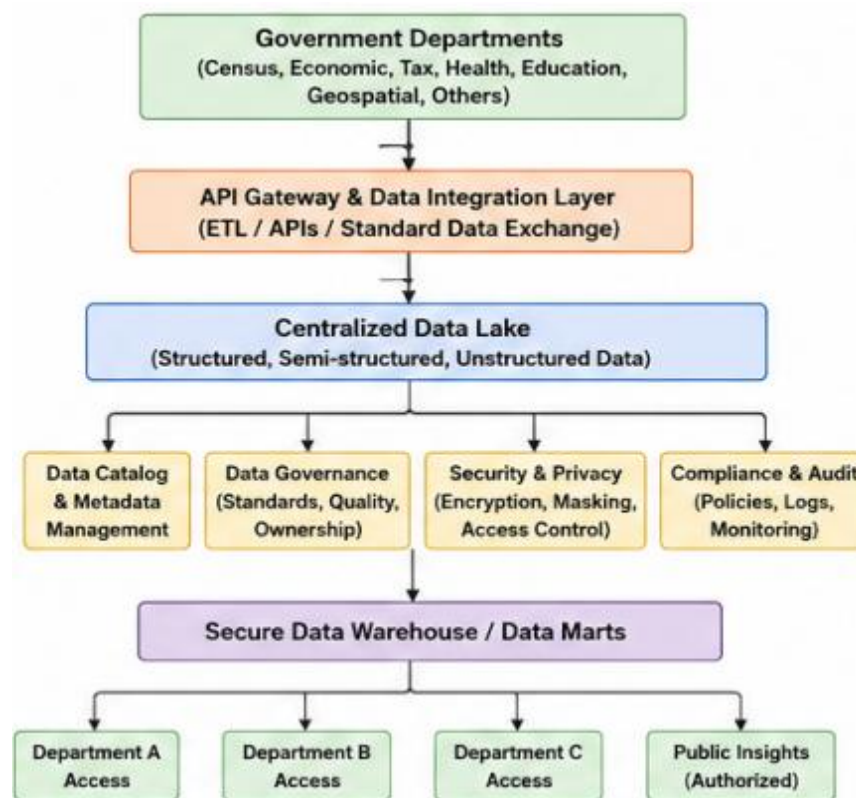
3. Government Data Platform Problem A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

A national government platform may integrate:

- Census data
- Economic data
- Tax information
- Geospatial data
- Healthcare data
- Education data
- Department databases
-

Different departments may use different formats and technologies. Therefore, the system requires interoperability, governance, security, and privacy.

Proposed Architecture



Data Integration: -

APIs and standardized data formats should be used to exchange information between departments.

Examples:

- REST APIs
- JSON
- XML
- Standard metadata
- Common identifiers

Data Governance: -

A centralized governance framework should define:

- Data ownership
- Data standards
- Data quality rules
- Access policies
- Metadata standards
- Data retention
- Data classification

Data Catalog: -

A data catalog maintains information about available datasets.

For example:

Dataset	Owner	Format	Sensitivity	Update
Census	Census Dept.	Structured	High	Annual
Economic	Finance Dept.	Structured	Medium	Monthly
GIS	Mapping Dept.	Geospatial	Medium	Daily

Security: -

Security mechanisms should include:

- Role-Based Access Control
- Multi-factor authentication
- Encryption
- API security
- Network segmentation
- Audit logs

Privacy: -

Sensitive citizen information should be protected through:

- Data masking
- Anonymization
- Pseudonymization
- Data minimization
- Controlled access

Interoperability: -

A common data model and standardized APIs allow departments to exchange information efficiently.

Benefits: -

- Single source of trusted data
- Reduced duplication
- Better interdepartmental collaboration
- Improved data quality
- Secure information sharing
- Better government decision-making

A government data platform should combine data lakes, APIs, metadata catalogs, common data standards, strong governance, encryption, and access controls to achieve secure and interoperable national data management.

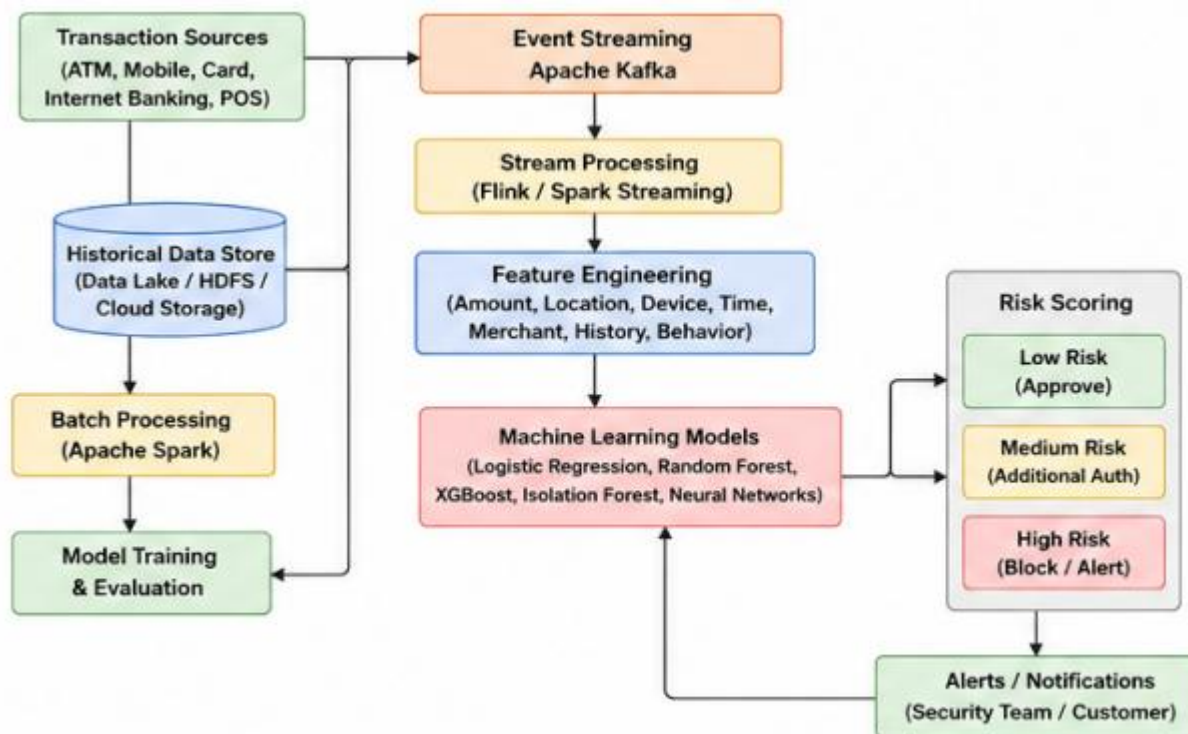
4. Banking	Fraud	Detection	Problem
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.			

Banks process millions of transactions every minute. Fraud detection requires:

- Real-time processing
- Very low latency
- High accuracy
- Historical analysis
- Continuous monitoring

A batch-only system is insufficient because fraud must be detected before or immediately after a transaction occurs.

Proposed Big Data Pipeline: -



Historical Data

Historical transaction data is stored in a data lake or distributed database.

Apache Spark can process historical data to:

- Identify fraud patterns
- Generate customer behavior profiles
- Train machine learning models

Real-Time Processing

Kafka receives transaction events.

Flink or Spark Streaming processes them with low latency.

Feature Engineering

Important features include:

- Transaction amount
- Transaction frequency
- Location
- Device ID
- Merchant
- Time
- Previous transaction behavior
- Number of failed attempts

Example:

Normal transaction:

₹2,000 → Chennai → Known device

Suspicious transaction:

₹1,50,000 → New country → Unknown device

The second transaction receives a high-risk score.

Algorithms: -

1. Logistic Regression

Useful for binary fraud classification.

2. Random Forest

Handles multiple transaction features effectively.

3. XGBoost

Provides strong predictive performance for structured transaction data.

4. Isolation Forest

Useful for detecting unusual transactions.

5. Neural Networks

Can identify complex nonlinear fraud patterns.

Benefits: -

- Real-time fraud detection
- Low response latency
- High detection accuracy
- Reduced false positives
- Continuous learning
- Ability to analyze historical and streaming data

A banking fraud detection system using Kafka, Flink/Spark, distributed storage, feature engineering, and machine learning can analyze millions of transactions in real time while maintaining high accuracy and low latency.

5. Agriculture	Smart	Farming	Problem
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.			

Smart farming generates data from multiple sources:

- Soil sensors
- Weather stations
- Drones
- Satellites
- IoT devices
- Irrigation systems
- Crop monitoring systems

The goal is to improve:

- Crop yield
- Water usage
- Fertilizer usage
- Pest management
- Farm productivity

Data Collection: -

Sensors continuously collect:

- Soil moisture
- Temperature
- Humidity
- Soil nutrients
- Rainfall
- Crop health

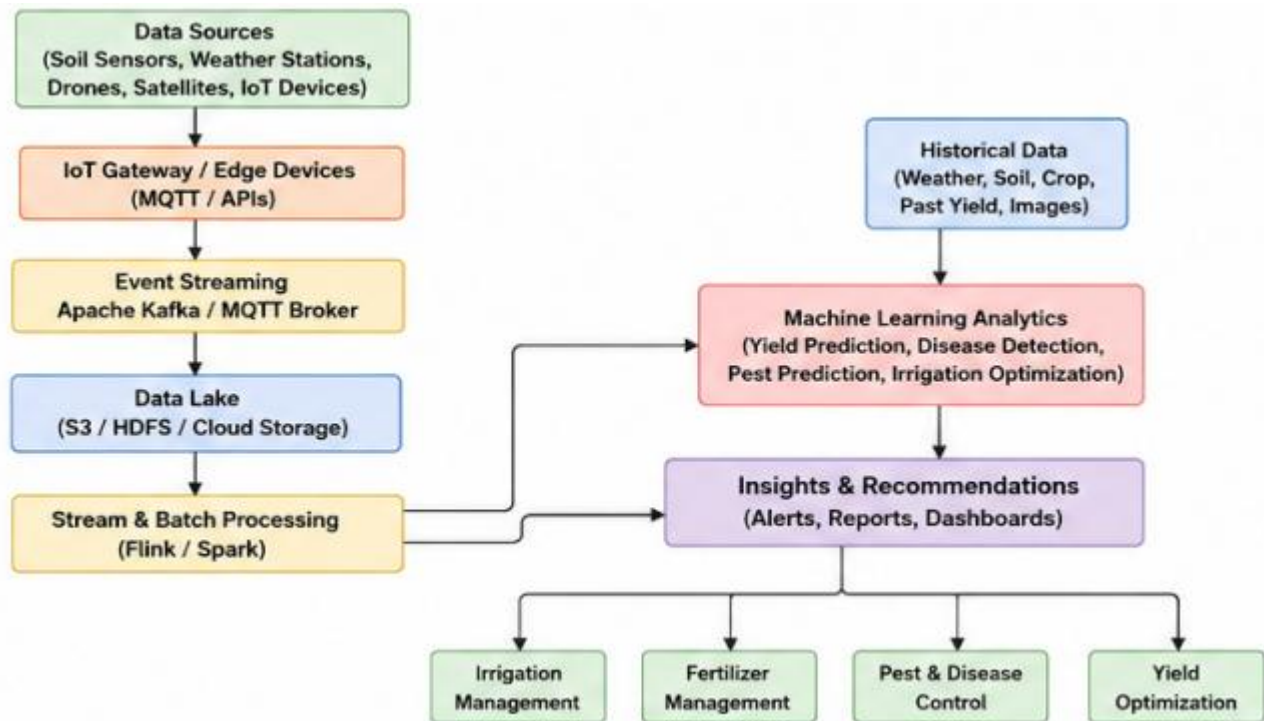
Drones can capture high-resolution images of fields.

Data Storage: -

Cloud object storage or distributed data lakes can store:

- Sensor data
- Drone images
- Satellite images
- Weather records
- Historical crop information

Proposed Big Data Architecture



Real-Time Analytics: -

Streaming systems can detect immediate conditions.

Machine Learning Applications: -

1. Yield Prediction

Models can predict crop yield using:

- Historical yield
- Soil conditions
- Weather

- Crop type
- Fertilizer usage

Algorithms:

- Random Forest
- XGBoost
- Neural Networks

2. Disease Detection

Drone and satellite images can be analyzed using computer vision and deep learning to identify unhealthy crops.

3. Irrigation Optimization

The system can determine how much water is required based on:

- Soil moisture
- Weather
- Crop type
- Temperature
- Rain forecast

4. Pest Prediction

Historical and environmental data can be used to predict potential pest outbreaks.

Benefits: -

- Increased crop yield
- Reduced water consumption
- Reduced fertilizer usage
- Early disease detection
- Better resource utilization

A precision agriculture solution using IoT sensors, Kafka/MQTT, cloud data lakes, Spark/Flink, machine learning, and drone imagery analytics can transform farming from traditional decision-making into data-driven smart agriculture.

